Topic: Transforming derivatives

Question: Find the Laplace transforms of $y'(t)$ and $y''(t)$, given $y(0) = 0$ and $y'(0) = 0$.

Answer choices:

A $\mathscr{L}(y'(t)) = s^2Y(s)$ $\mathscr{L}(y''(t)) = 0$

B $\mathscr{L}(y'(t)) = sY(s)$ $\mathscr{L}(y''(t)) = 2sY(s)$

C $\mathscr{L}(y'(t)) = sY(s)$ $\mathscr{L}(y''(t)) = s^2Y(s)$

D $\mathscr{L}(y'(t)) = 0$ $\mathscr{L}(y''(t)) = sY(s)$

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Solution: C

We'll start by rewriting the formula for the Laplace transform of a first derivative by replacing the function $f'(t)$ with the derivative function we were given, $y'(t)$. This will also change the transform F to the transform Y, and the initial condition from $f(0)$ to $y(0)$.

$$\mathscr{L}(f'(t)) = sF(s) - f(0)$$

$$\mathscr{L}(y'(t)) = sY(s) - y(0)$$

Now we can make a substitution into the transform equation for the initial condition.

$$\mathscr{L}(y'(t)) = sY(s) - 0$$

$$\mathscr{L}(y'(t)) = sY(s)$$

Now we'll rewrite the formula for the Laplace transform of a second derivative so that it's in terms of y and t instead of f and t.

$$\mathscr{L}(f''(t)) = s^2F(s) - sf(0) - f'(0)$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - sy(0) - y'(0)$$

Then we'll substitute for the initial conditions.

$$\mathscr{L}(y''(t)) = s^2Y(s) - s(0) - 0$$

$$\mathscr{L}(y''(t)) = s^2Y(s)$$

So for any function $y(t)$ with the initial conditions $y(0) = 0$ and $y'(0) = 0$, the Laplace transforms of its first and second derivatives will be

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$$\mathscr{L}(y'(t)) = sY(s)$$

$$\mathscr{L}(y''(t)) = s^2Y(s)$$

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Topic: Transforming derivatives

Question: Find the Laplace transforms of $y'(t)$ and $y''(t)$, given $y(0) = 1/2$ and $y'(0) = 3/4$.

Answer choices:

A $\mathscr{L}(y'(t)) = sY(s) - e^{2t}$ $\mathscr{L}(y''(t)) = sY(s) - e^{2t} - e^{\frac{4}{3}t}$

B $\mathscr{L}(y'(t)) = sY(s) - \frac{1}{2}$ $\mathscr{L}(y''(t)) = s^2Y(s) - \frac{s}{2} - \frac{3}{4}$

C $\mathscr{L}(y'(t)) = 2sY(s)$ $\mathscr{L}(y''(t)) = \frac{3s}{4}Y(s) - \frac{1}{2}s$

D $\mathscr{L}(y'(t)) = 2sY(s) - \frac{1}{2}$ $\mathscr{L}(y''(t)) = 2sY(s) - \frac{3}{4}s$

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Solution: B

We'll start by rewriting the formula for the Laplace transform of a first derivative by replacing the function $f'(t)$ with the derivative function we were given, $y'(t)$. This will also change the transform F to the transform Y, and the initial condition from $f(0)$ to $y(0)$.

$$\mathscr{L}(f'(t)) = sF(s) - f(0)$$

$$\mathscr{L}(y'(t)) = sY(s) - y(0)$$

Now we can make a substitution into the transform equation for the initial condition.

$$\mathscr{L}(y'(t)) = sY(s) - \frac{1}{2}$$

Now we'll rewrite the formula for the Laplace transform of a second derivative so that it's in terms of y and t instead of f and t.

$$\mathscr{L}(f''(t)) = s^2F(s) - sf(0) - f'(0)$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - sy(0) - y'(0)$$

Then we'll substitute for the initial conditions.

$$\mathscr{L}(y''(t)) = s^2Y(s) - s\left(\frac{1}{2}\right) - \frac{3}{4}$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - \frac{s}{2} - \frac{3}{4}$$

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So for any function $y(t)$ with the initial conditions $y(0) = 1/2$ and $y'(0) = 3/4$, the Laplace transforms of its first and second derivatives will be

$$\mathscr{L}(y'(t)) = sY(s) - \frac{1}{2}$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - \frac{s}{2} - \frac{3}{4}$$

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Topic: Transforming derivatives

Question: Find the Laplace transforms of $y'(t)$ and $y''(t)$, given $y(0) = \pi$ and $y'(0) = e$.

Answer choices:

A $\mathscr{L}(y'(t)) = sY(s) - e^{\pi}$ $\mathscr{L}(y''(t)) = s^2Y(s) - e^{\pi}s - (\pi + e)$

B $\mathscr{L}(y'(t)) = sY(s) - e^{\pi t}$ $\mathscr{L}(y''(t)) = s^2Y(s) - e^{\pi t} - e^{e}$

C $\mathscr{L}(y'(t)) = sY(s) - \pi$ $\mathscr{L}(y''(t)) = s^2Y(s) - \pi s - e$

D $\mathscr{L}(y'(t)) = sY(s) - e$ $\mathscr{L}(y''(t)) = s^2Y(s) - se - \pi$

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Solution: C

We'll start by rewriting the formula for the Laplace transform of a first derivative by replacing the function $f'(t)$ with the derivative function we were given, $y'(t)$. This will also change the transform F to the transform Y, and the initial condition from $f(0)$ to $y(0)$.

$$\mathscr{L}(f'(t)) = sF(s) - f(0)$$

$$\mathscr{L}(y'(t)) = sY(s) - y(0)$$

Now we can make a substitution into the transform equation for the initial condition.

$$\mathscr{L}(y'(t)) = sY(s) - \pi$$

Now we'll rewrite the formula for the Laplace transform of a second derivative so that it's in terms of y and t instead of f and t.

$$\mathscr{L}(f''(t)) = s^2F(s) - sf(0) - f'(0)$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - sy(0) - y'(0)$$

Then we'll substitute for the initial conditions.

$$\mathscr{L}(y''(t)) = s^2Y(s) - s\,(\pi) - e$$

$$\mathscr{L}(y''(t)) = s^2Y(s) - \pi s - e$$

So for any function $y(t)$ with the initial conditions $y(0) = \pi$ and $y'(0) = e$, the Laplace transforms of its first and second derivatives will be

$$\mathscr{L}(y'(t)) = sY(s) - \pi$$

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$$\mathscr{L}(y''(t)) = s^2Y(s) - \pi s - e$$

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